

Calculator & Test Strategy



Choosing between graphing and algebra

- 1 A system is given by $y = x^2 - 4x + 3$ and $y = 2x - 6$. Explain how graphing both functions on a calculator can quickly reveal the number of solutions, then find the solutions algebraically.
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Using a calculator to check algebraic work

- 2 A student solves $3x - 7 = 2x + 5$ and gets $x = 12$. Verify this solution by substitution, and explain why checking answers this way is a valuable test-taking strategy.
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Estimating and eliminating unreasonable answer choices

- 3 A multiple-choice question asks for the value of $\sqrt{50}$, with choices A) 5, B) 7.07, C) 25, D) 100. Without a calculator, use estimation to eliminate unreasonable choices, then find the exact simplified value.
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Efficient use of the calculator for systems of equations

- 4 Explain two different calculator-based strategies (other than the built-in equation solver) for solving the system $2x + 3y = 12$ and $x - y = 1$, and then solve it algebraically to confirm.
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Time management: identifying quick wins vs. time sinks

- 5 Compare the effort needed to solve $x + 5 = 12$ versus $3x^2 - 2x - 8 = 0$. Explain a time-management strategy for approaching a timed test section with a mix of question difficulties.
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Using calculator tables to explore function behavior

- 6 A function is given by $f(x) = x^2 - 6x + 5$. Using a table of values at $x = 0, 1, 2, 3, 4, 5, 6$, identify the zeros of the function and the approximate location of its minimum.
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Plugging in answer choices strategically

- 7 A multiple-choice question asks to solve $2x^2 - x - 6 = 0$, with choices A) $x = -1.5, 2$, B) $x = 1.5, -2$, C) $x = 3, -2$, D) $x = -1.5, -2$. Test choice C by substitution to determine if it's correct.
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Substituting numbers for variables in abstract problems

- 8 A problem states: if a is a positive integer, which expression is always even: $2a + 1$, a^2 , or $4a$? Test with $a = 1$ and $a = 2$ to determine the answer.
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Working backward from answer choices in word problems

- 9 A word problem states that a number increased by 40% equals 84, with answer choices A) 50, B) 60, C) 70, D) 84. Use the answer choices to work backward and find the correct original number.
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Deciding when a graph is faster than algebra

- 10 A question asks for the number of solutions to $|x - 3| = 2x + 1$ without asking for the actual solution values. Explain how graphing $y = |x - 3|$ and $y = 2x + 1$ answers this faster than solving algebraically, then confirm algebraically.
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Recognizing when an exact answer is required vs. an estimate

- 11 A question asks for the exact value of $\sqrt{72}$ in simplest radical form, while a related question asks for a decimal approximation to the nearest tenth. Provide both, and explain why a calculator alone is insufficient for the first part.
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A multi-part strategy problem: combining estimation, elimination, and verification

- 12 This question has three parts. A word problem states: "A rectangular garden has a perimeter of 46 feet and an area of 120 square feet. Find its dimensions." Answer choices are A) 10 by 12, B) 8 by 15, C) 6 by 17, D) 5 by 18. (a) Use the perimeter condition alone to eliminate choices quickly by estimation. (b) Check the remaining choices against the area condition. (c) State the correct dimensions and verify both conditions algebraically.
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Avoiding common sign and order-of-operations errors

- 13** A student evaluates -3^2 and gets 9, but the correct answer is -9 . Explain the order-of-operations error, then correctly evaluate -3^2 and $(-3)^2$ to show the difference.